# Preliminary Analysis

## 1. Introduction

An individual's mortgage application will result in either an approved or a denied loan, and that outcome is crucial in determining whether the individual can purchase a house. Under the Home Mortgage Disclosure Act (HMDA), lenders must submit the results of their applications, together with applicant, loan and property data. However, the files do not contain all of the factors the lender may have considered when making its decision.

Lewis-Faupel and Tenev (2024) analyzed applications submitted to U.S. lenders to find racial and ethnic differences in approval and pricing. In their abstract, they state that *"credit score, income and down payments each contribute significantly to disparities in mortgage access and affordability."* This finding motivated us to compare unadjusted denial rates with adjusted denial rates: if financial factors explain part of the gap between groups, the gap should change once those factors are taken into account.

### 1.1 Motivation

Unequal access to mortgages is a real-world problem, especially in countries with diverse racial groups. Access to a mortgage is one of the ways people are able to buy homes, and being denied a mortgage, or having to pay more interest, can affect a person's finances. Today, mortgage applications are evaluated both by automated systems and by human workers, and both can show biases. In this analysis we want to see whether acceptance or denial is associated with race, and whether other factors such as income relate to the issuance and denial of mortgages. We are also interested in how loan amount, loan type, debt-to-income ratio, lender and metro area interplay with race, and how these affect the decision to grant or deny a mortgage loan.

### 1.2 Research questions

1. Among 2025 home-purchase applications at three large lenders in New York, how do denial rates differ by reported applicant race, ethnicity, and the minority population percentage of the property's census tract?
2. How do those differences change when we account for income, loan amount, loan type, debt-to-income ratio, lender, and metro area? Do different combinations of these factors change the results?
3. Across the variables we measured, which are most strongly associated with denial, and which show the least association?

### 1.3 Expectations

By comparing denial rates together with group sizes and confidence intervals, we expected Black and Hispanic applicants to be denied more often than White applicants, and Asian applicants to fall between those groups: denied more often than White applicants but less often than Black and Hispanic applicants. We expected denial rates to rise with the minority share of the census tract. We also expected large groups, such as White applicants, to have narrow confidence intervals that support firmer conclusions.

Once we use logistic regression to control for income, loan amount, debt-to-income ratio, loan type, lender and metro area, we expect the racial gaps to shrink but remain statistically significant, mostly for Black and Hispanic applicants. We expect income and debt-to-income ratio to explain the largest share of the differences.

## 2. Data

### 2.1 Source and collection

We found this data while searching for public data on mortgages. It comes from the Home Mortgage Disclosure Act (HMDA) data published by the Federal Financial Institutions Examination Council (FFIEC). HMDA requires lenders to report nearly every mortgage application. We downloaded the 2025 New York applications for three large lending institutions. The downloaded file (hmda_loan_data.csv) contains 40,734 rows and 99 columns. Every row has an activity year of 2025 and a state code of NY.

The dataset is useful because it links many variables to each lending decision: applicant demographics, finances, loan details and neighborhood characteristics. This gives us a broad scope for analysis.

### 2.2 Data preparation and cleaning

We kept only the three outcomes that represent an actual decision by the lender on the application: originated (1), approved but not accepted (2), and denied (3). Purchased loans were excluded because the decision was made elsewhere, and withdrawn or incomplete applications were excluded because they produce neither an approval nor a denial. We then kept only home-purchase loans (loan_purpose = 1) with a first lien (lien_status = 1), for a principal residence (occupancy_type = 1), on a single-family (1–4 unit) site-built dwelling. Business or commercial loans and open-end lines of credit were excluded. This gives a smaller but more comparable sample, so our results only reflect decision-making on this subset of applications.

After these filters, the cleaned data contains 10,513 applications: 8,878 originated, 159 approved but not accepted, and 1,476 denied.

We also kept the 17 original fields listed in the codebook and converted income, loan amount, tract minority population percent, tract-to-MSA income percentage and tract population to numeric types. Blank, “NA” and “Exempt” values were treated as missing. Missing demographic information and unusual income values were kept in the cleaned file rather than dropped.

We created four variables:

- `denied`: the outcome variable, equal to 1 if action_taken = 3 and 0 if action_taken is 1 or 2. Denial records the decision made on the application. It does not indicate why the applicant was denied, and it cannot by itself demonstrate discriminatory behavior.
- `age_band`: a copy of applicant_age with the code 8888 relabeled as “Missing”.
- `dti_band`: HMDA reports debt-to-income ratio in a mixed format, mostly as ranges (e.g., “20%-<30%”) but as exact whole numbers from 36 to 49. We combined the values 36–49 into a single “36%-49%” band, giving six bands (<20%, 20%-<30%, 30%-<36%, 36%-49%, 50%-60%, >60%) plus “Missing.” For plotting, dti_band was set as an ordered categorical from lowest to highest.
- `tract_minority_band`: tract_minority_population_percent grouped into five non-overlapping 20-point bands (0–<20%, 20–<40%, 40–<60%, 60–<80%, 80–100%). This describes the neighborhood where the property is located, not the identity of the applicant.

**Missing and unusual values.** Income is recorded in thousands of dollars (173 = $173,000). In the cleaned data, 395 incomes are missing and 74 are zero or negative (10 are negative, which HMDA allows). Debt-to-income ratio is missing for 23 applications. For tract_to_msa_income_percentage, 62 rows are 0, which likely means the value is not available.

**Race groups used in comparisons.** For the race comparisons, we excluded “Race Not Available” and “Free Form Text Only” because they do not identify a group. We also excluded groups with fewer than 100 applications (American Indian or Alaska Native, n = 28; Native Hawaiian or Other Pacific Islander, n = 25; 2 or more minority races, n = 21), because percentages based on so few applications are too unstable to interpret. This leaves four groups: White, Asian, Black or African American, and Joint. Leaving out the smaller groups is a limitation of this analysis, not evidence that those groups experience no disparities.

The cleaned data is saved as hmda_2025_ny_cleaned.csv.

## 2.3 Description of variables

Table 1 summarizes the variables in the cleaned dataset. The full codebook is included in the project repository.

**Table 2. Variables in the cleaned dataset (10,513 applications, 21 columns)**

| Variable | Type | Description |
|---|---|---|
| denied (created) | Binary 0/1 | Outcome: 1 = denied, 0 = approved (originated or approved but not accepted) |
| derived_race | Categorical | Applicant race derived from applicant and co-applicant; "Joint" = applicant and co-applicant of different races, one of them White |
| derived_ethnicity | Categorical | Hispanic or Latino; Not Hispanic or Latino; Joint; Free Form Text Only; Not Available |
| applicant_age / age_band | Ordered categorical | <25 to >74; 8888 relabeled as Missing |
| income | Numeric | Gross annual income relied on by the lender, in $1,000s |
| debt_to_income_ratio / dti_band | Ordered categorical | Monthly debt payments divided by monthly income, grouped into bands |
| loan_amount | Numeric | Amount applied for, in dollars, reported at the midpoint of $10,000 intervals |
| loan_type | Categorical | 1 = Conventional; 2 = FHA-insured; 3 = VA-guaranteed |
| lei | Categorical | Lender's Legal Entity Identifier (3 lenders) |
| derived_msa-md | Categorical | 5-digit metro area / metro division code; 99999 = not in a metro area |
| county_code / census_tract | Categorical (geographic ID) | FIPS codes for county (5 digits) and census tract (11 digits) |
| tract_minority_population_percent / tract_minority_band | Numeric (0–100) / banded | Minority share of tract residents |
| tract_to_msa_income_percentage | Numeric | Tract median family income as % of metro median (100 = parity) |
| tract_population | Discrete count | Total tract residents |

# 3. Methods

## 3.1 Comparing denial rates across groups

Our first method is a direct comparison of denial rates. For each group of a categorical variable we computed the number of applications and the mean of denied, which is the share of applications denied. We did this for race, ethnicity, age band, DTI band, loan type and tract minority band. Reporting the group size alongside each rate matters because a rate based on a few dozen applications can be very different from the true rate just by chance.

For tract minority share, we also estimated 95% confidence intervals using seaborn's barplot, which uses bootstrapping: it resampled the data with replacement 1,000 times and took the 2.5th and 97.5th percentiles of the resampled denial rates. We chose this method because it gives an

interval for each group without needing extra assumptions, and it shows whether differences between neighboring groups are larger than the sampling uncertainty. We plan to add the same intervals to the race and ethnicity comparisons.

We chose descriptive comparison as the first step because our first research question asks how denial rates differ between groups before anything is controlled for. These unadjusted rates are the baseline that the adjusted models will be compared to.

### 3.2 Summary statistics and visualizations

Table 3 shows summary statistics for the numeric variables in the cleaned data.

**Table 3. Summary statistics for numeric variables**

| Statistic | denied | income ($1,000s) | loan_amount ($) | tract minority % | tract-to-MSA income % | tract population |
|---|---|---|---|---|---|---|
| Count | 10,513 | 10,118 | 10,513 | 10,513 | 10,513 | 10,513 |
| Mean | 0.140 | 290.7 | 669,450 | 42.9 | 143.6 | 4,847 |
| Std. dev. | 0.347 | 483.3 | 692,452 | 25.6 | 72.1 | 1,929 |
| Min | 0 | -719 | 15,000 | 4.17 | 0 | 44 |
| 25% | 0 | 106 | 265,000 | 22.52 | 92 | 3,497 |
| Median | 0 | 173 | 485,000 | 36.24 | 122 | 4,628 |
| 75% | 0 | 317 | 795,000 | 60.22 | 181 | 5,950 |
| Max | 1 | 16,328 | 9,595,000 | 99.78 | 292 | 13,385 |

Income has a long right tail: the median is roughly $173,000 but the mean is roughly $291,000, and the maximum is over $16 million. Because the largest incomes were too large to show on a linear scale, Figure 1 plots income on a log10 scale. Only the 10,044 positive incomes are plotted; the 469 missing or non-positive values remain in the cleaned data.

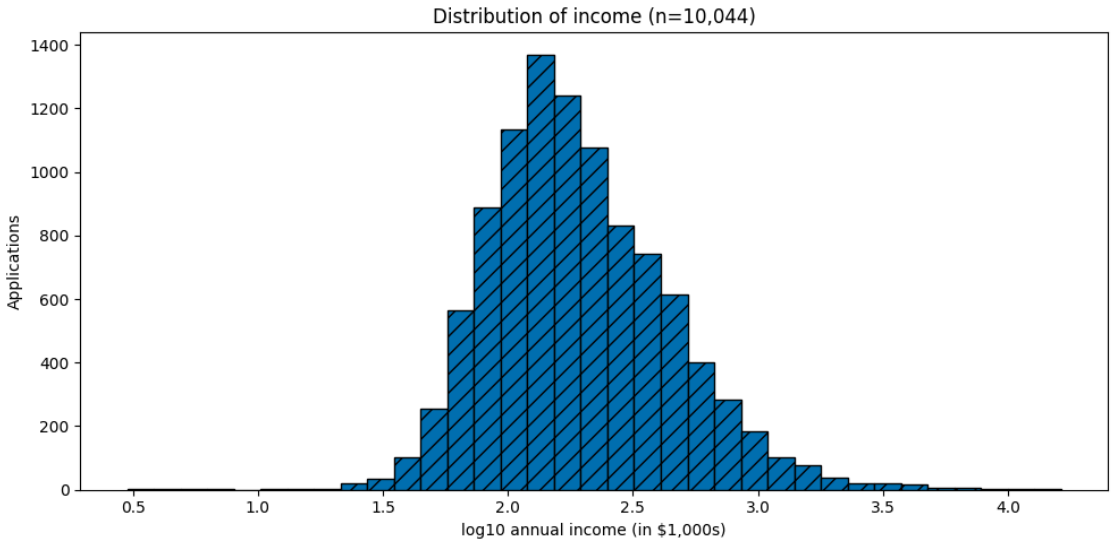



**Figure 1. Distribution of income (log10 scale, positive incomes only, n = 10,044)**

*Alt text: Logarithmic histogram of annual income in thousands of dollars, including only positive values. Bars are blue with black outlines and diagonal hatching so they can be identified in grayscale. Most applications fall between about 10^1.8 and 10^2.8, that is roughly $60,000 to $600,000, with a peak near 10^2.15 (about $140,000) and a long tail to the right.*

Because debt-to-income ratio is a major consideration in lending decisions, it could explain part of any difference in denial rates between groups. Figure 2 shows the distribution of DTI bands within each race group, as a 100% stacked bar chart. Percentages are calculated within each race so groups of different sizes can be compared directly. A scatter plot was not suitable because both variables are categorical.

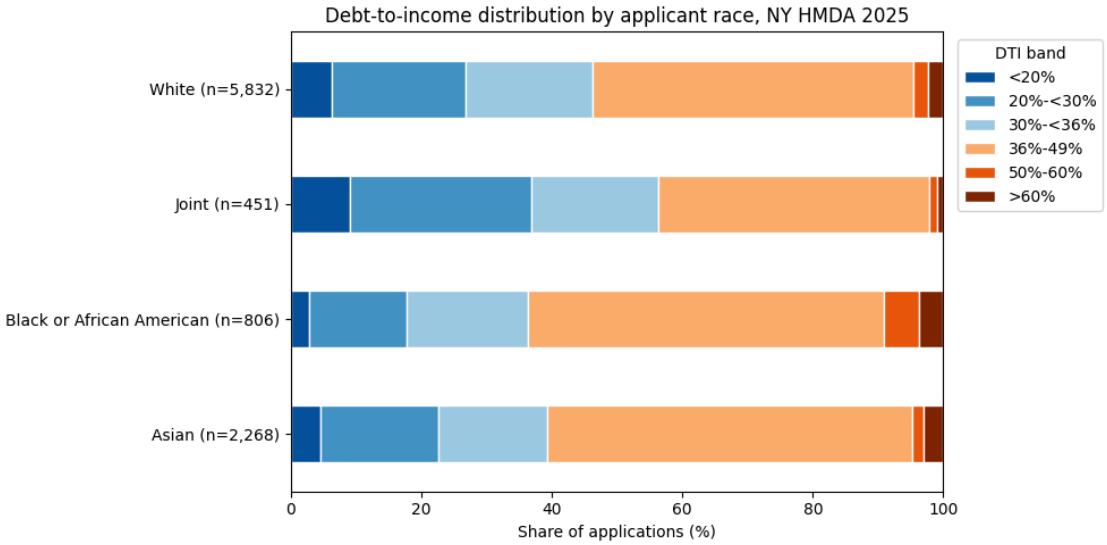



**Figure 2. Debt-to-income distribution by applicant race, NY HMDA 2025**

*Alt text: Horizontal 100% stacked bar chart of debt-to-income bands by applicant race. Colors run from dark blue (lowest DTI) to dark brown (highest DTI). White (n=5,832): about 46% under 36% DTI and about 4% above 50%. Joint (n=451): about 56% under 36% and about 2% above 50%. Black or African American (n=806): about 36%*

*under 36% and about 9% above 50%. Asian (n=2,268): about 40% under 36% and about 5% above 50%. The 36% to 49% band is the largest for every group.*

For every group, the 36%–49% band is the largest. Black applicants have the largest share of high-DTI applications, with about 9% above 50% DTI compared with roughly 2% to 5% for the other groups. Joint applicants have the most low-DTI applications, with about 56% under 36%. (The group sizes in Figure 2 are slightly smaller than in Table 4 because applications with missing DTI were excluded from this chart.)

## 3.3 Modelling

### Logistic regression

To test whether racial and ethnic gaps in denial remain after accounting for other factors, we fit logistic regression models with denied as the outcome, using statsmodels. The race models include White, Black and Asian applicants with positive income, with White as the baseline. Income and loan amount were log-transformed because both are highly right-skewed. We fit the race models in two parts:

- Part 1: race only, then race plus applicant controls (log income, log loan amount, DTI band, age band, loan type and lender).
- Part 2: race plus tract minority share (per 10 points) and metro area, then the same model with the applicant controls added.

Ethnicity was modelled separately, comparing Hispanic and non-Hispanic applicants, because it overlaps heavily with race: most Hispanic applicants are classified as White, and only 82 are Black and 16 Asian. The ethnicity models go in three steps: ethnicity only, then adding tract minority share and metro area, then adding the applicant controls. Results are reported as odds ratios with 95% confidence intervals and p-values; an odds ratio above 1 means higher odds of denial than the baseline group.

### Explainable Boosting Model

To assess whether there were underlying relationships between our predictive variables and denial based on EBM's ability to detect these beyond what a standard logistic model would identify, we fit an Explainable Boosting Model (EBM). Our predictor variables were; Income, Loan Amount, Tract Minority Share, Race/Ethnicity, DTI Band, Loan Type, Lender, Metro Area. The number of observations was kept at 10,513 as all data points from our original dataset were kept after cleaning, which included replacement of any missing or nonpositive incomes in both

sets with the median value from the training sample and addition of dummy variables for nonpositive income and unavailable income.

The data was then randomly divided into 80% training set (8410 records), 20% test set (2103 records) using a stratified sampling approach to preserve the denial proportion of the full dataset. A random seed of 42 was applied to ensure reproducibility. The model was fit using version 0.7.8 of InterpretML, with four outer bags, 5000 max iterations, internal early stopping and no interaction terms allowed so that we could evaluate each feature independently. Test set performance of the model was assessed via three metrics: ROC-AUC, Log Loss, Brier Score compared against a naive baseline that assigns every applicant the training denial rate. Performance of the model was also assessed by evaluating classification accuracy at a fixed probability cut-off of .5. In order to understand how well the model is fitting, selected relationship plots were created and ranking of features by average absolute contribution to model output are provided.

## 4. Results

Of the 10,513 applications in the cleaned data, 1,476 were denied, an overall denial rate of 14.0%.

### 4.1 Denial rates by race

**Table 4. Denial rate by applicant race (* fewer than 100 applications; not interpreted)**

| derived_race | Applications | Denial rate |
|---|---|---|
| White | 5,841 | 11.8% |
| Asian | 2,271 | 15.1% |
| Black or African American | 808 | 24.6% |
| Joint | 452 | 11.5% |
| Race Not Available | 1,056 | 16.6% |
| American Indian or Alaska Native* | 28 | 21.4% |
| Native Hawaiian or Other Pacific Islander* | 25 | 12.0% |
| 2 or more minority races* | 21 | 23.8% |
| Free Form Text Only* | 11 | 27.3% |

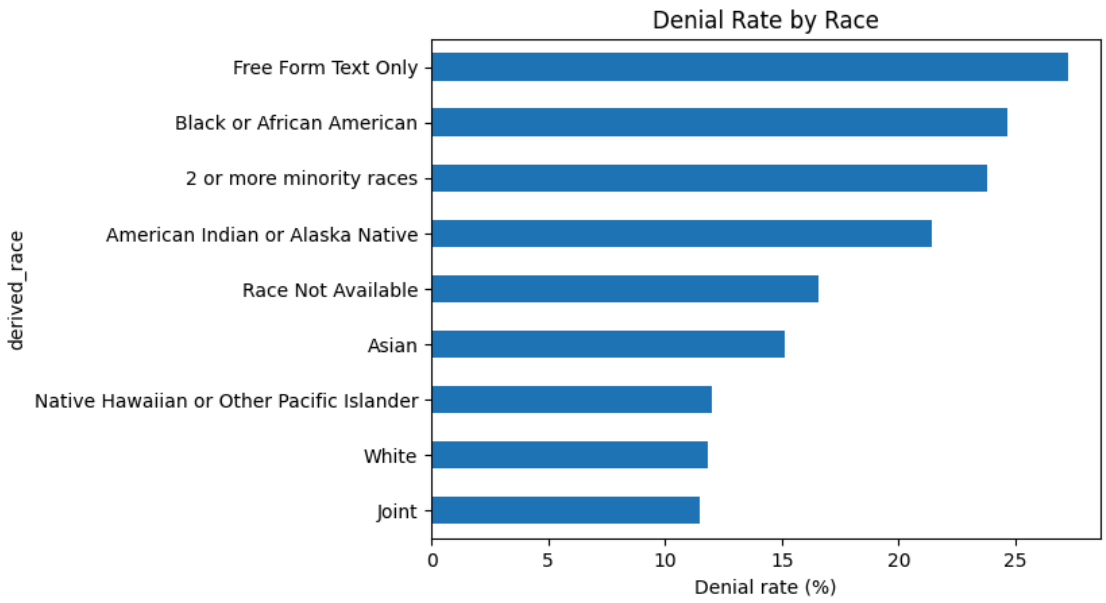



**Figure 3. Denial rate by race, NY HMDA 2025**

*Alt text: Horizontal bar chart of denial rate (%) by derived race, sorted from highest to lowest: Free Form Text Only 27.3%, Black or African American 24.6%, 2 or more minority races 23.8%, American Indian or Alaska Native 21.4%, Race Not Available 16.6%, Asian 15.1%, Native Hawaiian or Other Pacific Islander 12.0%, White 11.8%, Joint 11.5%. Several of the top bars are based on very few applications.*

Among the large groups, Black applicants are denied most often, at 24.6%. That is about twice the rate for White applicants (11.8%), so a Black applicant in this data was roughly twice as likely to be turned down. Asian applicants are in between, at 15.1%. Joint applicants, where one applicant is White and the co-applicant is another race, have the lowest rate at 11.5%, about the same as White applicants. Applicants who did not report their race (Race Not Available) are denied 16.6% of the time.

Some of the highest bars in Figure 3, such as Free Form Text Only at 27.3%, are misleading because those groups have only 11 to 28 applications each. This is why they are excluded from our main comparisons.

## 4.2 Denial rates by ethnicity

**Table 5. Denial rate by applicant ethnicity**

| derived_ethnicity | Applications | Denial rate |
|---|---|---|
| Not Hispanic or Latino | 8,408 | 13.2% |
| Hispanic or Latino | 934 | 21.8% |
| Joint | 326 | 9.5% |
| Ethnicity Not Available | 831 | 15.3% |
| Free Form Text Only | 14 | 14.3% |

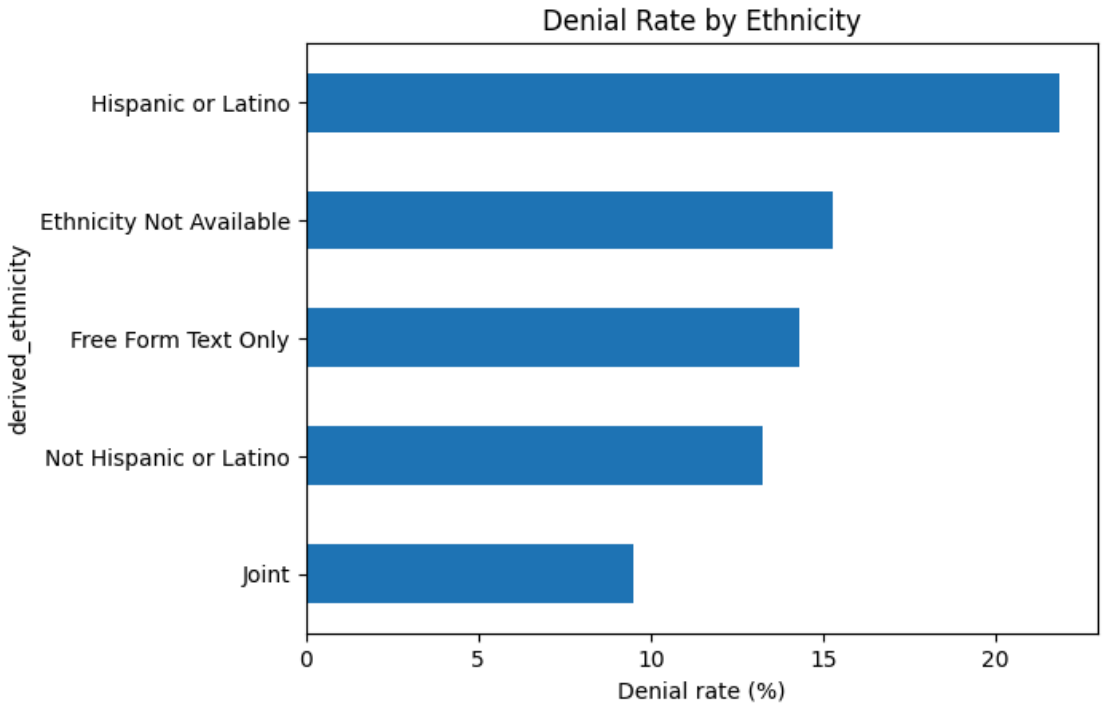



**Figure 4. Denial rate by ethnicity, NY HMDA 2025**

*Alt text: Horizontal bar chart of denial rate (%) by derived ethnicity, sorted from highest to lowest: Hispanic or Latino 21.8%, Ethnicity Not Available 15.3%, Free Form Text Only 14.3%, Not Hispanic or Latino 13.2%, Joint 9.5%.*

Hispanic or Latino applicants are denied 21.8% of the time, compared with 13.2% for Not Hispanic or Latino applicants, about 1.7 times as often. Joint applicants, with one Hispanic and one non-Hispanic applicant, have the lowest rate at 9.5%.

## 4.3 Denial rates by tract minority share

**Table 6. Denial rate by minority population share of the property's census tract**

| Tract minority share | Applications | Denial rate |
|---|---|---|
| 0–<20% | 2,005 | 11.0% |
| 20–<40% | 3,804 | 11.7% |
| 40–<60% | 2,055 | 13.1% |
| 60–<80% | 1,287 | 15.2% |
| 80–100% | 1,362 | 25.4% |

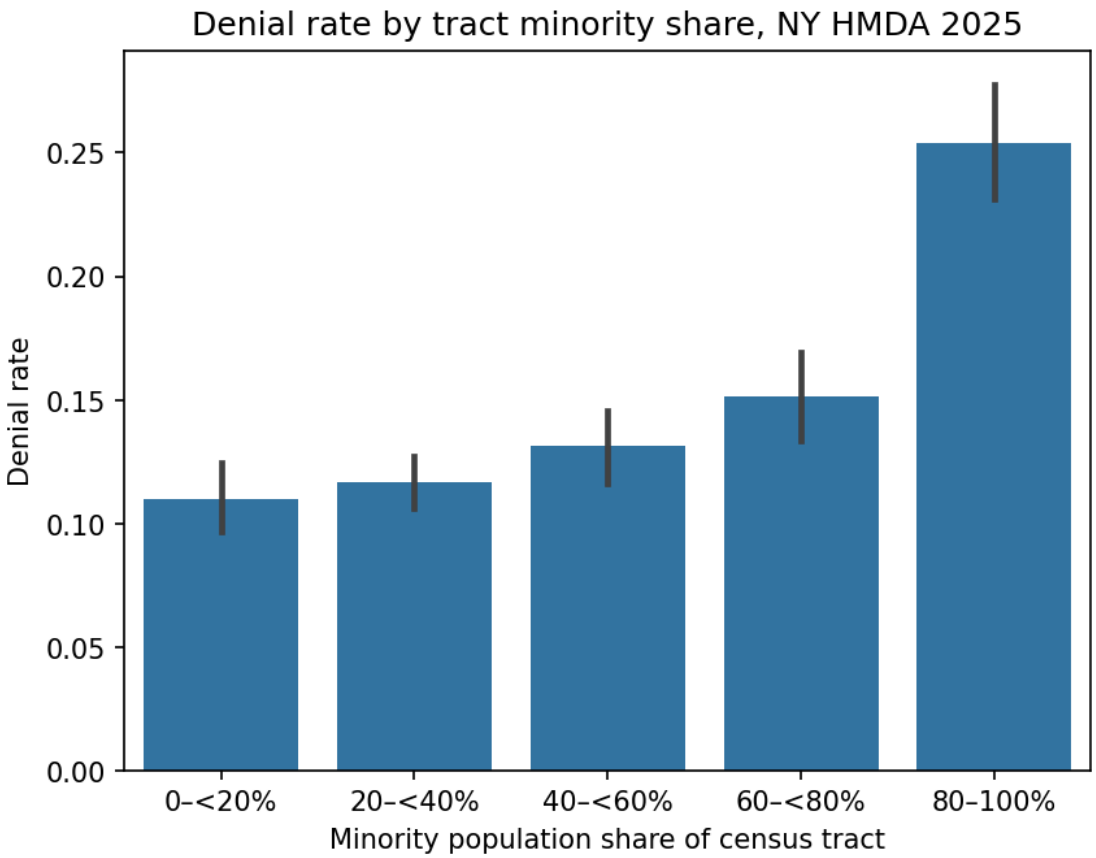



**Figure 5. Denial rate by tract minority share, with 95% bootstrap confidence intervals**

*Alt text: Bar chart of mortgage denial rates by the minority population share of the property's census tract, NY HMDA 2025. Error bars show 95% confidence intervals. Denial rates rise gradually from about 11% in tracts under 20% minority to about 15% in tracts 60% to 80% minority, then jump to about 25% in tracts 80% to 100% minority, whose error bar does not overlap any other band.*

Denial rates rise gradually across the first four bands, from about 11% in tracts under 20% minority to about 15% in tracts that are 60–80% minority. The confidence intervals of neighboring bands overlap in this range, so the differences between adjacent bands are modest. The 80–100% band stands apart: its denial rate of 25.4% is more than double that of the lowest band, and its confidence interval does not overlap with any other band.

### 4.4 Other variables associated with denial

As a first look at the third research question, Table 7 shows denial rates for the remaining categorical variables.

**Table 7. Denial rate by DTI band, loan type and age band**

| Variable | Group | Applications | Denial rate |
|---|---|---|---|
| DTI band | <20% | 605 | 11.6% |
| | 20%-<30% | 2,111 | 8.8% |

| Variable | Group | Applications | Denial rate |
|---|---|---|---|
| | 30%-<36% | 1,946 | 8.9% |
| | 36%-49% | 5,276 | 10.5% |
| | 50%-60% | 264 | 74.6% |
| | >60% | 288 | 96.9% |
| | Missing | 23 | 82.6% |
| Loan type | 1 – Conventional | 10,285 | 13.4% |
| | 2 – FHA-insured | 177 | 49.2% |
| | 3 – VA-guaranteed | 51 | 15.7% |
| Age band | <25 | 251 | 21.5% |
| | 25-34 | 3,305 | 12.1% |
| | 35-44 | 3,573 | 12.7% |
| | 45-54 | 1,777 | 16.7% |
| | 55-64 | 1,029 | 17.2% |
| | 65-74 | 456 | 17.8% |
| | >74 | 111 | 14.4% |
| | Missing | 11 | 9.1% |

Debt-to-income ratio shows the strongest association with denial of any variable we examined. Applications below 50% DTI are denied about 9% to 12% of the time, but the rate jumps to 74.6% for the 50%–60% band and 96.9% for applications above 60%. Loan type also shows a large difference: FHA-insured applications are denied 49.2% of the time, compared with 13.4% for conventional loans, although only 177 FHA applications are in the sample. Age shows a smaller pattern: the youngest applicants (<25) have the highest denial rate at 21.5%, and rates among applicants 45 and older (14% to 18%) are somewhat higher than among those aged 25 to 44 (about 12% to 13%).

### 4.5 Explainable Boosting Machine results

The EBM was able to achieve a ROC-AUC of .791, a log loss of .283 and a Brier score of .076 on 2,103 test applications. The baseline that used the training denial rate for all applications had a log loss of .405 and a Brier score of .121. Using a probability threshold of .5 the EBM was correct on 91.2% of applications, however it only found 120 out of the 295 denied applications, therefore the model's denial recall was 40.7%. As such, while this represents high overall accuracy, it also indicates that there were 175 denials that the model failed to predict.

These results represent the results from only one exploratory data split and, without comparable logistic regression test results, provide no basis to conclude if the use of EBM produces better predictive ability than logistic regression.

The DTI band contributed the most to the EBM's prediction with respect to the average absolute contribution, followed by loan amount, metro area, and lender. Income ranked #7. While these rankings indicate how heavily the fitted model relies upon each variable, they do not determine causal relationships nor demonstrate statistical significance.

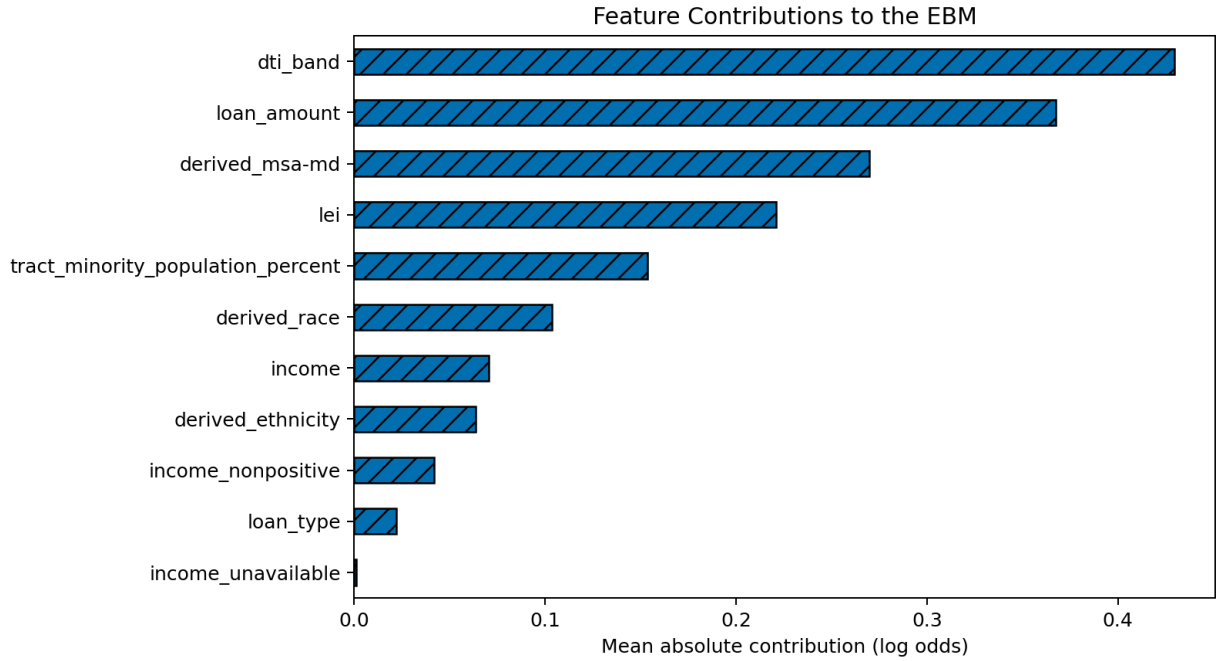



**Figure 6. EBM feature importance by mean absolute contribution to model's score**

*Alt text: Horizontal bars rank the features in descending order according to their mean absolute contribution to the fit in log odds units. A longer bar indicates that this particular feature has a greater contribution than another shorter bar for this specific model.*

The income contribution to the predicted denial generally decreased through the middle of its range, and then became uneven at higher income levels, where there were fewer applications.

Tract minority share tended to have a greater positive contribution to the EBM's prediction of denial as you move up the tract minority share range. These contributions are expressed in log odds and therefore do not directly translate into percentage point increases in denial probability. Additionally, since there is no indication of uncertainty ranges for the plots presented above, we do not consider small fluctuations or "jumps" as representative of definitive thresholds.

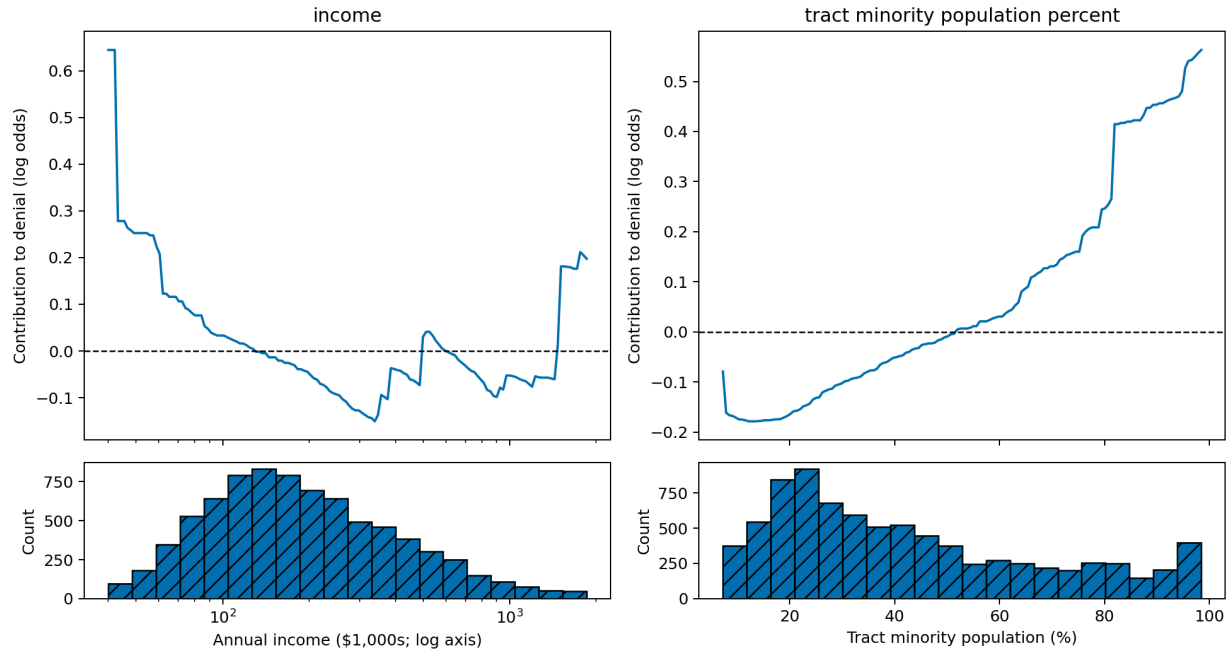
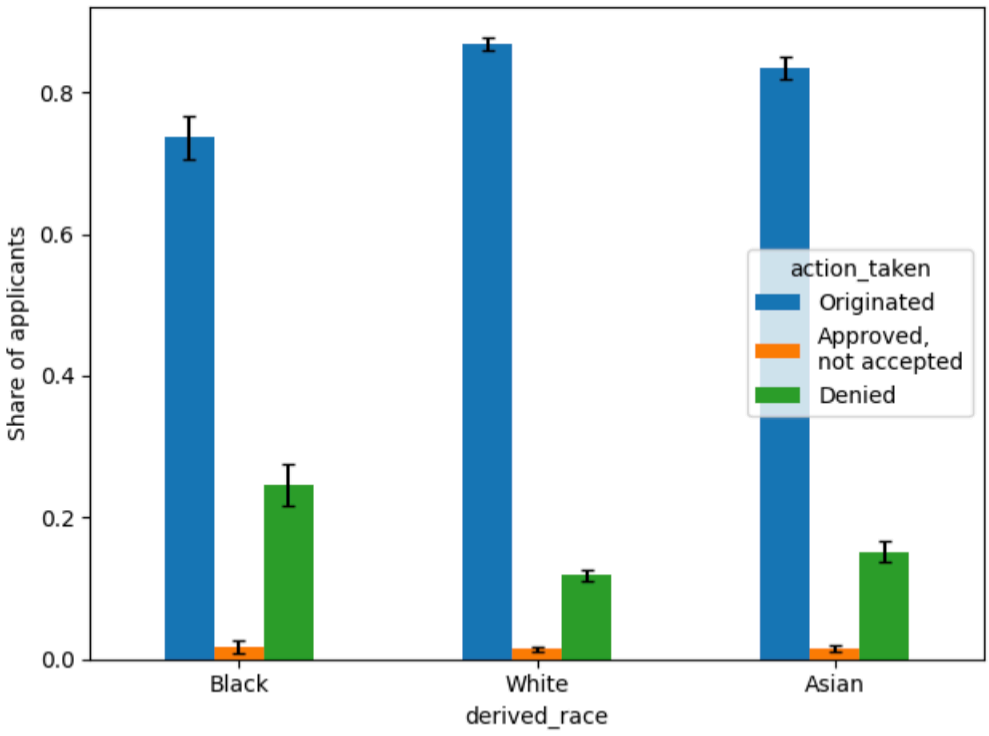



**Figure 7. EBM feature contributions for income and tract minority share, with training-data distributions beneath**

*Alt text: Four panels show EBM feature contributions above their respective training data distributions. Income's contribution to estimated denial generally declines through the mid-range, but it fluctuates as income levels rise, especially at the highest income levels (where there are fewer applications). A logarithmic scale for income is shown in thousand dollar increments. The contribution of the percent of a tract that is composed of minorities tends to increase as the percent minority share of a tract increases. The increase becomes steeper at about 80% minority.*

## 4.6 Logistic Regression results

Across all three groups, most applications originated, and approved-but-not-accepted was rare (about 1% to 2%). Black applicants had the largest denied share (about 25%), versus about 15% for Asian and 12% for White applicants. We expected more Black applicants given New York's population, yet there were only 808 of them versus 2,271 Asian and 5,841 White applicants. The graph shows a gap but cannot show why it exists. This makes us to move in the logistic regression to control for certain features




| action_taken | Originated | Approved,\nnot accepted | Denied |
|---|---|---|---|
| derived_race | | | |
| Black | 0.736 | 0.017 | 0.246 |
| White | 0.868 | 0.014 | 0.118 |
| Asian | 0.835 | 0.014 | 0.151 |

**Figure 8. Application outcomes by applicant race, NY HMDA 2025**

*Alt Text: Grouped bar chart of application outcomes for Black, White and Asian applicants, with 95% confidence interval error bars. Each group has three bars: originated (blue), approved but not accepted (orange) and denied (green). Originated is 73.6% for Black, 86.8% for White and 83.5% for Asian applicants. Denied is 24.6% for Black, 11.8% for White and 15.1% for Asian applicants. Approved but not accepted is about 1% to 2% in every group. The denied error bars do not overlap between groups.*

Using White as the baseline:

Black applicants were denied at roughly twice (2.45) the White rate, and even after controlling for income, loan amount, DTI, age, loan type and lender, they had about 1.74 times the odds of denial.

Asians were 1.32 times more likely to be denied than the White which is moderate compared to the rates I saw with Asians. After controlling for income, loan amount, DTI, age, loan type and lender, they had about 1.47 times the odds of denial. This is a rise from the 1.32 which is surprising. All four estimates are statistically significant. I expected controls to shrink both gaps, but only the Black gap shrank.

```
Race only:
                                        odds_ratio  ci_low  ci_high
C(race, Treatment('White'))[T.Asian]          1.32    1.15     1.53
C(race, Treatment('White'))[T.Black]          2.45    2.04     2.94

With controls:
                                        odds_ratio  ci_low  ci_high
C(race, Treatment('White'))[T.Asian]          1.47    1.23     1.74
C(race, Treatment('White'))[T.Black]          1.74    1.39     2.18
```

**Figure 9. Odds ratios of mortgage denial for Hispanic vs. non-Hispanic applicants, NY HMDA 2025**

*Alt Text: Three side-by-side bar charts showing the share of applicants by outcome (Originated, Approved but not accepted, Denied) for Black (n=808), White (n=5,841) and Asian (n=2,271) applicants. Originated is the tallest bar in every panel. The Denied bar is tallest for Black applicants at about 25%, versus about 15% for Asian and 12% for White applicants.*

I analyzed ethnicity separately instead of adding it to the race model because Hispanic ethnicity overlaps heavily with race in this data (most Hispanic applicants are classified as White, with only 82 Black and 16 Asian Hispanic applicants)

Hispanic applicants had 1.73 times the odds of denial of non-Hispanic applicants. Accounting for tract minority share and metro area lowered this to 1.43, and adding applicant controls lowered it to 1.16, which is no longer statistically distinguishable from 1. Higher tract minority share stayed linked to higher denial odds in both models, about 11% to 15% per 10 points. I expected the ethnicity gap to shrink with controls, and it did.

```
1. Ethnicity only:
           odds_ratio  ci_low  ci_high
hispanic         1.73    1.42     2.11

2. + tract minority % and MSA:
                     odds_ratio  ci_low  ci_high
hispanic                   1.43    1.17     1.75
tract_minority_10          1.15    1.12     1.18

3. + applicant controls:
                     odds_ratio  ci_low  ci_high
hispanic                   1.16    0.92     1.48
tract_minority_10          1.11    1.08     1.15
```

**Figure 10. Black applicants were denied about twice as often as White applicants, NY HMDA 2025**

*Alt Text: Table of odds ratios for mortgage denial from three logistic regression models comparing Hispanic to non-Hispanic applicants. Rows are Hispanic ethnicity, tract minority share per 10 points, and the number of applications. Columns give each model's odds ratio, 95% confidence interval and p-value. Hispanic ethnicity falls from 1.73 in Model 1 to 1.43 in Model 2 and 1.16 in Model 3, where the interval includes 1. Tract minority share is 1.15 in Model 2 and 1.11 in Model 3, with p-values below 0.001.*

Adding neighborhood and metro area, Black applicants had 1.68 times the odds of denial of White applicants. Adding applicant controls cut that to 1.29, and the interval (1.01 to 1.64) barely excludes 1. Asian applicants' odds were not clearly different from White applicants' in either model. Each 10-point rise in tract minority share was linked to about 10% higher odds of denial. I expected the Black gap to shrink with controls, and it did, though it did not disappear.

```
1. Race + tract minority % + MSA:
                     odds_ratio  ci_low  ci_high
race: Asian                1.12    0.96     1.30
race: Black                1.68    1.37     2.07
tract_minority_10          1.13    1.10     1.16

2. + applicant controls:
                     odds_ratio  ci_low  ci_high
race: Asian                1.17    0.98     1.41
race: Black                1.29    1.01     1.64
tract_minority_10          1.10    1.06     1.14

Rows used: 8529
```

**Figure 11. Odds ratios of mortgage denial by race and tract minority share, NY HMDA 2025**

*Alt Text: Table of odds ratios for mortgage denial from two logistic regression models, with White applicants as the baseline. Rows are Black applicants, Asian applicants and tract minority share per 10 points. Columns give each model's odds ratio, 95% confidence interval and p-value. Black applicants' odds ratio falls from 1.68 in Model 1 to 1.29 in Model 2 and stays above 1. Asian applicants' intervals include 1 in both models. Tract minority share is 1.13 in Model 1 and 1.10 in Model 2, with p-values below 0.001.*

# 5. Discussion

## 5.1 Summary

Among 2025 home-purchase applications at three large New York lenders, unadjusted denial rates differ noticeably by race, ethnicity and neighborhood. Black applicants were denied about twice as often as White applicants (24.6% vs. 11.8%), Hispanic or Latino applicants about 1.7 times as often as Not Hispanic or Latino applicants (21.8% vs. 13.2%), and applications for properties in tracts that are 80–100% minority were denied about 25% of the time, more than double the rate in tracts under 20% minority. Among the other variables, debt-to-income ratio is by far the most strongly associated with denial, followed by loan amount, metro area and lender.

Descriptive comparisons allow us to address our first research question, while regressions and EBMs expand on this by allowing us to examine the specific contribution of each variable in the model as a whole. The EBM also shows that tract minority share contribution was increasing at the higher end of the distribution; therefore, we can say that neighborhood composition continued to be associated with predicted denial within the context of the fitted model. This is an extension of the simple neighborhood comparison results, but it does not necessarily mean that lenders were using neighborhood racial composition as a component in their lending decision-making process. In terms of how much each of these factors contributes to the predictions made by the EBM (our third question), DTI, Loan Amount, Metro Area and Lender were all top contributors. Income ranked 7th, thus providing less support for our expectation that income would have been one of the strongest predictors. These rankings provide information about how much each factor contributes to the model's score and do not establish why lenders are making the decisions they make.

## 5.2 Connection to the literature

Lewis-Faupel and Tenev (2024) found racial and ethnic differences in mortgage approval and pricing, and reported that credit score, income and down payments each contribute significantly to those disparities. Our unadjusted results are consistent with the existence of racial and ethnic differences in denial. Our finding that DTI is very strongly associated with denial, and that Black applicants have a larger share of high-DTI applications, is consistent with their point that financial factors contribute to disparities.

## 5.3 Limitations and critique of our methods

- Several race groups (American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, 2 or more minority races) had fewer than 100 applications and were excluded.
- Combining DTI values 36 through 49 into one band makes it the largest group and hides how denial rates vary across each integer value within it.
- Applications with missing race or ethnicity ("Not Available") are excluded from the main comparisons, and they have above-average denial rates (16.6% and 15.3%).
- Some categories are small, such as FHA (177) and VA (51) loans, so their rates are less reliable.

### 5.4 Reliability, validity and limits on inference

HMDA data is reported by lenders under a federal requirement, which makes it a broad and consistent source. However, the public files do not contain all of the factors a lender may have considered. In particular, Lewis-Faupel and Tenev (2024) name credit score and down payment as significant contributors to disparities, and these are not among the variables in our dataset. The denied variable records the decision on the application; it does not tell us why the applicant was denied, and it cannot demonstrate discriminatory behavior.

The sample covers only three large lenders, only New York, only the year 2025, and only first-lien, owner-occupied, single-family home-purchase applications that ended in an approval or denial. Results may not generalize to other lenders, other states, other years, refinancing or other loan purposes, or to applications that were withdrawn or left incomplete. Because the data is observational, any association we find between race and denial, even after controlling for other variables, cannot on its own be interpreted as a causal effect of race, since factors not reported in HMDA may differ between groups.

### 5.5 Ethical concerns and societal implications

Access to a mortgage is one of the main ways people buy homes, so evidence about unequal access has real societal importance. A benefit of this work is that it uses public data to describe differences in how applicants of different races, ethnicities and neighborhoods are treated, which can help identify where closer examination is needed.

There are also risks. Because denial does not show why an application was denied and our data is missing factors such as credit score, there is a risk of over-interpreting raw gaps as proof of discrimination, or, after controlling for other variables, of wrongly concluding that no unfair treatment exists. Excluding small race groups and applicants with missing race means our results say nothing about those applicants. Tract minority share describes a neighborhood, not an

individual applicant, so neighborhood results should not be read as statements about any individual. We aim to present our results with these limitations clearly stated.

## References

Federal Financial Institutions Examination Council (FFIEC). Home Mortgage Disclosure Act (HMDA) data, 2025, New York.

Lewis-Faupel and Tenev (2024).